



AEO “Nazarbayev Intellectual Schools”

CHEMISTRY

Grade 12

Paper 2

May 2024

MARK SCHEME

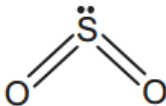
Maximum Mark: 100

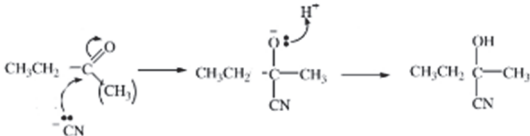
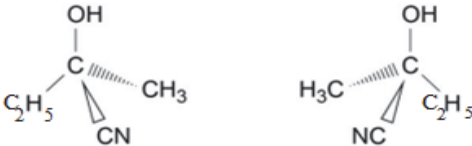
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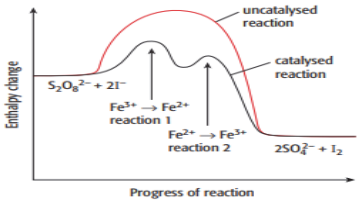
ANNOTATIONS AND ABBREVIATIONS

line space	new marking point for same question item
/	meaning “or” indicates alternative correct responses for same marking point
() brackets	the words or units in brackets do not need to be stated but must be implied For example, 10 (J) means that the mark is awarded for “10” but not for “10 kJ”
<u>underlining</u>	exact word is required
accept	Indicates a less than perfect, but in this case, acceptable response.
allow	identifies an answer that will be allowed if no other marking point is awarded.
not	Indicates an incorrect response that would contradict another otherwise correct alternative i.e. right plus wrong penalty applies.
ignore	Indicates an incorrect response which is irrelevant and does not cause a right plus wrong penalty.
ecf	meaning "error carried forward", where a wrong response in an earlier part of a question is used correctly in a later part of the same question
sig. figs	Answers are normally acceptable to any number of significant figures ≥ 2 . This indicates an exception where a specified number of significant figures is required. Numbers must be correctly rounded and not truncated.
note	often added for in additional guidance to help the exam marker.
con	this is used when marking exam papers to indicate that a learner has given a contradictory answer and loses a mark.
ora	or reverse argument
owtte	or words to that effect
AVP	any valid point

PRE-STANDARDISATION

Question	Answer	Mark	Additional guidance
1(a) (i)	$2\text{PbS} + 3\text{O}_2 = 2\text{PbO} + 2\text{SO}_2$	[1]	
1(a) (ii)	$\text{moles} = \frac{2000}{239.3} = 8.36$ $\text{moles}(\text{PbS}) = \text{moles}(\text{SO}_2) = 8.36$ $V = 8.36 \times 24 = 200 \text{ dm}^3 \text{ or } 0.2 \text{ m}^3$	1 1 [max 2]	ecf is considered
1(b) (i)	 -bent shape e pair -bond angle <120°	1 1 1 [max 3]	Accept diagram with 4 bonds and 1 lone pair Accept non-linear shape Accept 115°-120°
1(b) (ii)	SiO₂ - giant covalent lattice with strong covalent bonds SO₂ – simple molecular/ simple covalent with weak intermolecular forces.	1 1 [max 2]	Accept network structure macromolecule
1(c) (i)	- low enough to achieve an acceptable yield - high enough to do this in an acceptable time	1 1 [max 2]	Accept -high for sensible rate optimum temperature for catalyst Accept some comment on yield/ rate and compromise required
1(c) (ii)	$K = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2[\text{O}_2]}$	[1]	square brackets essential ignore any units
1(c) (iii)	Kc remains constant/ does not change. An idea that an equilibrium is disturbed and after restored.	1 1 [max 2]	Accept owtte

Question	Answer	Mark	Additional guidance
2(a) (i)	Butan-2-one/ butanone	1	Ignore -If the structure of X only given
	From orange to green	1 [max 2]	- green ppt Accept green Cr ³⁺ solution
2(a)(ii)	Nucleophilic addition  both curly arrows dipole correctly shown curly arrow from lone pair on O⁻ to H⁺ completely correct structure of product	1	NOT accept
		1	wrong direction of curly arrows;
		1	if HCN is used instead of CN-
		1	
		[max 5]	
2(b)(i)		1	NOT Accept
		1	If enantiomers drawn as 2D
		[max 2]	
2(b) (ii)	an idea that enantiomers rotate plane of polarized for different direction	[1]	OWTTE Accept diagram
2(b)(iii)	Advantages:	1	OWTTE
	- prevent side-effects - less dosage of drug - higher potency Disadvantages: -manufacturing of pure enantiomer can be expensive - difficult to separate pure enantiomer from racemic mixture	1 [max 2]	

Question	Answer	Mark	Additional guidance
3(a) (i)	gain of electrons.	[1]	Accept decrease in the oxidation state gain of hydrogen
3(a) (ii)	$E^\ominus_{\text{cell}} = 2.01 - 0.54 = 1.47 \text{ V}$	[1]	
3(a) (iii)	Iodide, from -1 to 0	[1]	
3(b) (i)	 Curve for catalysed reaction must be shown in two steps Curve for uncatalysed reaction	1 1 [max 2]	
3(b) (ii)	Catalyzed reactions- positive iron ions and negative ions are attracted to each other Uncatalysed reaction-the peroxodisulfate and iodide ions both have a negative charge, need considerable energy to overcome the repulsive forces	1 1 [max 2]	Ignore lower Ea Accept individual stages involve collision between positive and negative ions. This will be much more likely to be successful than collision between two negative ions in the uncatalysed
3(c)(i)	variable oxidation state/ incomplete d sub-shells	[1]	Accept - ability to adsorb - ability to form complex ions
3(c)(ii)	pollution	[1]	Accept Any reasonable answer

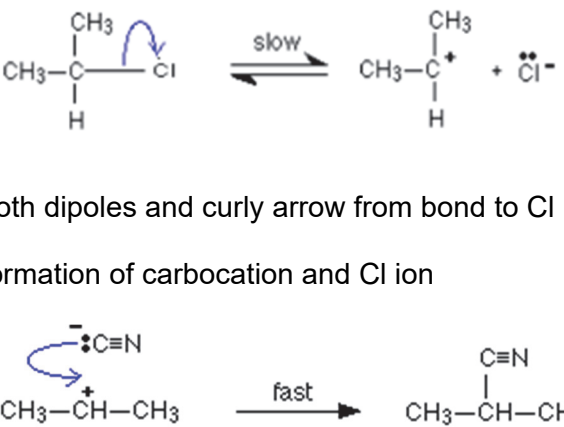
Question	Answer	Mark	Additional guidance
4a (i)	benzene has lower energy / more stable due to delocalisation pi system/ resonance energy	1 1 [max 2]	Accept diagram showing π bond Kekulé structure evolves $360 \text{ kJ mol}^{-1} / 3 \times 120$ $360 - 208 = 152 \text{ kJ}$
4a (ii)		1 1 1 [max 3]	1 mark for each arrow
4a (iii)	$\Delta H_f + (-3267) = 6(-394) + 3(-286)$ $\Delta H_f[\text{C}_6\text{H}_6(\text{l})] = 3 \cdot (-286) + 6 \cdot (-394) - (-3267) = 45 \text{ kJ mol}^{-1}$	1 1 [max 2]	AWARD 2 marks for correct answer IGNORE Units NOT accept -45 kJ mol^{-1}
4b (i)	A - $\text{PCl}_5/\text{SOCl}_2/\text{PCl}_3$ C -	1 1 [max 2]	Accept names Accept $\text{COOC}_2\text{H}_5 / \text{COOCH}_2\text{CH}_3$ for ester group
4b (ii)	<i>Any two from:</i> Compound B produces hydrogen chloride and not water reacts faster more vigorously reacts with alcohols without the need for a catalyst or H^+ more reactive produces higher yield reacts irreversibly	1 1 [max 2]	ora
4c (i)	6	[1]	Accept Shifts or labelling

4c (ii)		[1]	Accept -molecular formula
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Question	Answer	Mark	Additional guidance
5(a) (i)	Carboxylic group	[1]	Accept -COOH
5(a) (ii)	G-C three hydrogen bonds A-T two hydrogen bond	1 1 [max 2]	
5(b)	-molecules very similar to the substrate molecules bind to the active site -prevent binding of the actual substrate	1 1 [max 2]	Accept bonding with active site
5(c) (i)	HA + H ₂ O ⇌ H ₃ O ⁺ + A ⁻ or HA ⇌ H ⁺ + A ⁻	1 [1]	Must be equilibrium sign
5(c) (ii)	HA (acid) and A ⁻ (conjugate base)/ H ₂ O (base) and H ₃ O ⁺ (conjugate acid)	1 [1]	
5(c) (iii)	$[H^+] = \sqrt{5} \times 10^{-5} \times 0.2 = 0.00316 \text{ mol dm}^{-3}$ $pH = -\log[H^+] = 2.5$	1 1 [max 2]	Accept ecf from first mark if working or evaluation of [H ⁺] is present ALLOW Correct answer with no working scores

Question	Answer	Mark	Additional guidance
6(a) (i)	Silicon	[1]	Accept Si
6(a) (ii)	More protons/greater nuclear charge/proton number increases	1	Ignore just 'increasing atomic number'
	Greater attraction between nucleus and (outer) electrons/(outer) electrons held more strongly by nucleus	1	Accept outer electrons in same shell(energy level) or decrease in atomic radius or attracting the same number of electron shells (energy level)
	Same(similar) shielding	1	
		[max 3]	
6(a) (iii)	for S one 3p orbital has paired electrons/ for P 3p sub-shell has unpaired electrons AND paired electrons repel/ spin pair repulsion	[1]	owtte
6(b) (i)	$\text{Mg(s)} + 2\text{H}_2\text{O(l)} = \text{Mg(OH)}_2\text{(aq)} + \text{H}_2\text{(g)}$	[1]	Correct equation with state symbols
6(b) (ii)	More effervescence/bubbles/react more vigorously	[1]	Accept more vigorous
6(b) (iii)	Solubility increase	1	Accept -become more soluble in water as you go down the column -lattice dissociation and hydration enthalpy become more exothermic/ more negative - lattice enthalpy decreases faster than hydration enthalpy
	Lattice and hydration enthalpies BOTH decrease / the radius of the metal cations increase	1	
	Lattice enthalpy increases faster than hydration enthalpy /with decreasing ionic size	1	
		[max 3]	
6(c) (i)	Neutralization/ acid-base reaction/ calcium oxide react with sulfur dioxide gas	[1]	Accept $\text{Ca(OH)}_2\text{(s)} + \text{SO}_2\text{(g)} \rightarrow \text{CaSO}_3\text{(s)} + \text{H}_2\text{O(l)}$
6(c) (ii)	Increase acidity/ decrease pH of soil and water/ erosion/ damaging of building	[1]	Accept damage to crops and vegetation/forest damage/metal corrosion/ effect to the biodiversity

Question	Answer	Mark	Additional guidance
7a (i)	$\text{CH}_3\text{-CH}_2\text{-CH}_3 + \text{Cl}_2 = \text{CH}_3\text{-CHCl-CH}_3 + \text{HCl}$	[1]	NOT accept Molecular formulas
7a (ii)	- elimination - Alcoholic/ ethanolic KOH	1 1 [max 2]	Ignore Heating/reflux
7b (i)	$ \begin{array}{cccccc} \text{CH}_3 & \text{H} & \text{CH}_3 & \text{H} & \text{CH}_3 & \text{H} \\ & & & & & \\ -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C}- \\ & & & & & \\ \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} \end{array} $	[1]	All bonds must be single Accept -correct structural OR displayed OR skeletal formula OR mixture of the above (as long as unambiguous) -CH ₃ and H above or below chain IGNORE any brackets drawn and n
7b (ii)	high strength	[1]	Accept Any from -chemical resistance -water resistance -heat resistance -electrical insulation -mechanical properties
7b (iii)	addition polymers are non-biodegradable Consist of strong C-C bonds/ covalent bonds	1 1 [max 2]	Accept are <u>inert</u> (unreactive) as they have strong C-C bonds. Have single carbon-carbon bonds and take many years to break down
7c (i)	(from expt 1 & 2:/ [CN ⁻] const) [C ₃ H ₇ Cl] doubles, rate doubles 1st order for C ₃ H ₇ Cl (from expt 1 & 3:/ [C ₃ H ₇ Cl] const) [CN ⁻] doubles, rate doubles 1st order wrt CN ⁻ Rate = k[CN ⁻][C ₃ H ₇ Cl]	1 1 1 [max 3]	
7c (ii)	rate of SN1 is only dependent on concentration of the substrate	1	Accept - rate of SN1 is only dependent on of the [C ₃ H ₇ Cl]

7c (iii)	 both dipoles and curly arrow from bond to Cl formation of carbocation and Cl ion CN⁻ attacks carbocation from lone pair Correct structure of product	Must be 2 steps NOT accept SN₂ mechanism 1 1 1 1 [max 4]
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Question	Answer	Mark	Additional guidance
8a (i)	resists changes in pH when small amount of acid or alkali is added.	[1]	owtte
8a (ii)	The bicarbonate buffer is a rapid response system, since the product of its interaction with CO₂ acids is quickly eliminated through the lungs. The pH is balanced by the presence of both a weak acid (eg H₂CO₃) and its conjugate base (eg HCO₃⁻), so that any excess acid or base entering the system is neutralized.	1 1 [max 2]	Accept Other reasonable answers
8b	$K_a = \frac{[H^+][HCO_3^-]}{[H_2CO_3]}$ $[H^+] = K_a \times \frac{[H_2CO_3]}{[HCO_3^-]}$ $[H^+] = 4.35 \times 10^{-7} \times \frac{0.125}{0.05} = 10.9 \times 10^{-7} / 10.87 \times 10^{-7}$ pH=6/5.96.	1 1 1 1 [max 4]	Accept 4.60 Must be to 2 DP if the concentrations find wrongly - lose 1 mark and next mark gained

Question	Answer	Mark	Additional guidance
9a (i)	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$	[1]	Accept [Ar] $4s^2 3d^7$
9a (ii)	3	[1]	
9b (i)	central metal ion surrounded by coordinately bonded ligands	[1]	Accept metal ion attached with a number of other molecules or ions by dative covalent bond
9b (ii)	$[\text{Co}(\text{H}_2\text{O})_6]^{2+} + 6\text{NH}_3 \longrightarrow [\text{Co}(\text{NH}_3)_6]^{2+} + 6\text{H}_2\text{O}$ For equation For balancing	1 1 [max 2]	Ignore Equilibrium sign
9b (iii)	EDTA is multidentate/ammonia is unidentate/ chelate effect $\Delta G = -\Delta H - T\Delta S$ Enthalpy slightly changes AND Entropy increases/more particles on RHS/more positive ΔG is more negative/ $\Delta G < 0$	1 1 1 1 [max 4]	
9c	Catalyze the conversion of carbon dioxide to hydrogencarbonate ions/ CO_2 to HCO_3^-	[1]	

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